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Depressive symptoms and associated factors among people living with HIV receiving antiretroviral therapy in Pokhara, Nepal: a cross-sectional study

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Abstract

Background:
Depression is a major global health problem that affects million of people. People Living with HIV (PLWH) are at higher risk of Depressive symptoms as compared to general population. Evidence from low- and middle-income countries show the prevalence of depressive symptoms ranging from 0% to 63.3%.

Objective:
To measure the prevalence of depressive symptoms and explore its determining factor among people living with HIV under antiretroviral therapy centre in Pokhara, Nepal.

Design:
Cross-sectional design was applied.

Setting:
This study was carried out in an Institutional based setting from February to March 2019 at ART centre of Pokhara Academy of Health Science, Pokhara Nepal.

Participant:
Total of 256 participants were included in face-to-face interviews, while 22 did not respond. Among 256 participant 141 were female and 115 were male

Primary and Secondary Outcomes Measure:
The Depression prevalence is the primary outcome which is assessed by Nepali version of Beck Depression Inventory (BDI) tool and 16/17 cut-off was used as recommended by Kohrt BA et al. The test of association of depression with different variables like Demographic, Economic and Socio-behavioural were secondary outcomes.

Results:
The prevalence of depressive symptoms was 19.1% (95% CI: 14.8–24.4%).The multivariable analysis showed, Divorced/separated/widowed participants had 3 times higher odds of depressive symptoms than married participants (AOR 3.05, 95% CI 1.29–7.35; p=0.012). Participants with a monthly income ≤10,000 NRP had 2.89 fold higher odds of depressive symptoms than those earning >10,000 NRP (AOR 2.89, 95% CI 1.24–7.12; p=0.017). Participant perceiving discrimination had 2.5 times higher odds of depressive symptoms (AOR 2.49, 95% CI 1.14–5.40; p=0.021) than those who did not, and participants with current tobacco consumption had 4.8 times higher odds of depressive symptoms than those who did not (AOR 4.84, 95% CI 2.04–12.1; p=0.001), whereas other variables like Sex, Ethnicity, Family Occupation, Drug Use in 12 months, and other variable were not statistically associated.

Conclusions:
The study concluded that approximately one-fifth of the PLWH in the ART centre had depressive symptoms using the BDI tool and Marital Status, Monthly income, Perceived discrimination and Tobacco consumption were associated factors of depressive symptoms in PLWH. The study findings highlight the burden of depressive symptoms and recommend the mental health and appropriate psychosocial support to PLWH in ART with considering risk associated factors. Furthermore, multicentre and longitudinal research is needed to better understand the relation between HIV and depression in PLWH.

Keywords: Depression, Depressive Symptoms, Depression in PLWH, PLWH Nepal

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Background/Rationale

Depression is a global mental health problem that presents with depressed mood, loss of interest, decreased energy, feeling of guilt or low self-worth, disturbed sleep or appetite and poor concentration (1) Depression affects in all aspects of life, including performance at school, productivity at work, relationships with family, friends and community. In 2021, 332 million people in the world had depression which including 5.7% of adults and 5.9% of adults aged 70 years and older (2).

According to the National Mental Health Survey of Nepal (2019/2020), the lifetime prevalence of any mental health disorder was 10%, while the current prevalence was 4.3%. The lifetime prevalence of major depressive disorder (MDD) was 3%, with a current prevalence of 1%, which is lower than global estimates (3). On the otherhand, depression is considerably more prevalent among people living with HIV (PLWH). Studies from low- and middle-income countries have reported prevalence estimates ranging from 0% to 63.3% (4), while in study from Nepal reported to prevalence from 25.5% to 34% (5,6). Overall, PLWH are higher risk of depressive symptoms than the general population, Overall, PLWH are at higher risk of depressive symptoms than the general population. Comprehensive strategies integrating pharmacological treatment, psychological support, and social interventions are therefore important for improving health outcomes the need for comprehensive strategies that integrate pharmacological treatment, psychological support, and social interventions to improve overall health outcome of PLWH(7) .

Nepal has made substantial progress in its HIV response through the transitioning the UNAIDS 90–90–90 strategy to the 95–95–95 targets under the Nepal HIV Strategic Plan 2021–2026 (8). In fiscal year 2018/2019 total 17,987 patients were enrolled in antiretroviral therapy throughout Nepal (9). By 2020, 83% of people living with HIV (PLWH) knew their status, 66% were receiving sustained ART, and 32% had achieved viral suppression, despite major health-system restructuring (8). However, the national HIV strategic plans do not explicitly integrate mental health services or interventions to address the psychological well-being of PLWH.

In a study from the Kathmandu Valley Nepal, high family support was associated with an 80% reduction in the odds of depressive symptoms among PLWH (AOR = 0.20, 95% CI: 0.08–0.53), while high internalized stigma was associated with more than 4 times higher odds of depressive symptom (AOR = 4.17, 95% CI = 1.76-9.83) (10). Similarly, another study from the Kathmandu Valley reported lower odds (AOR = 0.27, 95% CI: 0.12–0.63) of depressive symptoms among employed PLWH compared with unemployed PLWH, while drug use with 6 months was associated with higher odds of depressive symptom (AOR = 3.63, 95% CI: 1.23–10.73) than no use of drug.Adding on, high perceived family support was associated with reduced odds of depressive symptoms compared with low support (AOR = 0.25, 95% CI: 0.07–0.91) (6). Adding on, a study from Lumbini Province, Nepal, reported that literate PLWH had lower odds of depressive symptom compared with illiterate (AOR = 0.24, 95% CI: 0.10–0.61), whereas those in the poorest wealth quintile had significantly higher odds of depressive symptoms compared with those in the richest quintile (AOR = 7.28, 95% CI: 2.22–23.87) (10) .

Depression among PLWH is demographic, clinical, socioeconomic, behavioural, and psychosocial factors, including female sex, older age, high viral load, unemployment, low income, substance use, poor social support, and psychological vulnerabilities. The relation of HIV and Depression is bi-directional. HIV infection increases the risk of depression through complex biological and psycho-social mechanisms, while depression adversely affects HIV treatment and disease progression (11). Depression not only affects the health and well-being of PLWH but depression and HIV can have serious health implication and it has double burden of disease. When a depressed patient exhibits symptoms associated with risky behaviour on non-adherence to medication as well as may endanger both patient and others in the community (12). Although depression is recognized as a major comorbidity among PLWH, evidence from Nepal remains limited, particularly regarding its prevalence and associated factors at the local level. Such evidence is essential to inform the integration of mental health services into HIV care and support evidence-based program planning (13). And to our knowledge this first study assessing prevalence of depressive symptoms in PLWH in Kaski District. Therefore, the main objective of this study to determine the prevalence of depressive symptoms and identify factors associated with it among PLWH in Kaski District, Nepal.

This study may provide a baseline evidence which can helpful in development and implementation on mental health intervention for PLWH in Nepal.

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Methods:

Study design and Settings:

This study used cross-sectional study design done with in an institutional setting at ART centre of Pokhara Academy of Health Science (PAHS). The ART centre at Pokhara Academy of Health Science is largest ART centre in Gandaki province which give regular services to approximately 1000 patient out of 2165 (9). The community-based sampling was not feasible due to lack of baseline on community-level sampling frame of PLWH, therefore institutional based study was performed. This study involved quantitative interview with using socio-demographic questionnaire and a tool for depression assessment. The study population comprised PLWH enrolled at the ART centre who attended the centre for ART services or routine follow-up, the data collection period was between February and March 2019.

Study Participant

The eligibility participant were PLWH aged ≥18 years who provided written informed consent to participate. If a participant did not meet the eligibility criteria, the next eligible client on the ART centre was approached and invited to participate in the study.

Sample size and sampling procedure
The sample size was calculated using the single population proportion formula considering following details: since the prevalence was unknow, therefore 50% was prevalence was assumed. A 95% confidence level and a 5% margin of error were used. The total number of ART clients at the study clinic was approximately 1,000 (N=1000).

The sample size (n) for this study was determined by following equation:

$$n = \frac{NX}{N + X - 1}$$

Where,
X = initial sample size before finite population correction
$Z_{0.05}$ = Value of Z at α (0.05) = 1.96
p = prevalence of depressive symptoms in PLWH = 50%
p= 0.5
q = (1-p) = 0.5
e = level of error 5 % = 0.05

For calculation of value of X

$$X = \frac{{Z_{\alpha/2}}^2 * p * (1 - p)}{e^2}$$

so X= $\{(1.96)^2 \text{ X } 0.50 \text{ X } 0.50\} / (0.05)^2$
X = 385

Putting the value of X in sample size formula

$$(n) = \frac{NX}{N + X - 1}$$

=(1000*385)/(384+1000-1)
= 278

Therefore, the required sample size was 278 participants.

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Accordingly, 278 patients attending the ART clinic were approached during the study period.

Data sources/ measurement:

Data were collected using a structured questionnaire administered by trained interviewers through face to face interviews. The questionnaire collected information on sociodemographic characteristics and health-related behaviours. Depressive symptoms were assessed through 21 items Beck's Depression Inventory questionnaire (14).

A valid Nepali version of the 21-item Beck Depression Inventory (BDI) was used to assess depressive symptoms among PLWH. Each items was scored from 0 to 3, yielding a total score ranging from 0 to 63. We used the 16/17 cut-off as the recommendation by Kohrt BA et al, cut-off score of ≥17 (16/17) was used to identify participants with depressive symptom. This threshold demonstrated good diagnostic performance, with a sensitivity of 0.86 and a specificity of 0.85 and a Cronbach's alpha of 0.88. Although lower cut-off scores may improve sensitivity, they may reduce specificity and increase the likelihood of false-positive screening results (15,16). Following data collection, the completed questionnaires were coded, checked for completeness and consistency, and entered into IBM Statistical Package for the Social Sciences (SPSS) version 26 for data management. Statistical analyses were performed using RStudio.

The primary outcome variable was depressive symptoms, measured using the validated Nepali version of BDI and dichotomized into a binary variable (Yes/No) based on the predefined cut-off score.The explanatory variables included demographic characteristics (age, sex, ethnicity, residence, religion, marital status, educational status, employment status, monthly personal income, participant occupation, family occupation, and family income) and socio-behavioural factors (family support, spouse support, involvement in family events, perceived discrimination, current tobacco consumption, current alcohol consumption, drug use in the past 12 months, and use of medication for depression). We used the STROBE reporting guideline to draft included in supplement A (17).

Bias:
To handle selection bias, all eligible patients attending the ART Centre during the study period were approached for participation. Information bias was reduced by using the validated Nepali version of the Beck Depression Inventory (BDI), which ensured linguistic and cultural appropriateness. The questionnaire was pretested at Teku ART centre, Kathmandu before data collection to assess its clarity and usability, and the revisions of whole study proposal were made based on feedback and recommendations from the Nepal Health Research Council (NHRC) during the ethical review process.

Statistical methods:

At first, descriptive statistics was used to summarized the data, followed by test of association using chi-square and fisher's exact test was used for test of association and followed by Multivariable regression modelling. Continuous variables were described using measures of central tendency and dispersion, including mean, median, first quartile (Q1), and third quartile (Q3), while categorical variables were presented using frequencies and percentages. Due to the skewed distribution of monthly income and the absence of an established income threshold, the first quartile (Q1) was used as a data-driven cut-off to categorize monthly income into two groups. Age was categories into two categories below Q3 and above Q3 above. Missing values can be seen in two variable; occupation of participant and support of spouse variable which applies to employed participant and married participant.

Associations between categorical variables were assessed using Pearson's chi-square test or Fisher's exact test, where appropriate, and then multivariable logistic regression was used for adjustment of confounder.At first, variables with $p<0.05$ in the initial analyses were included in the initial model, however perceived spouse support and occupation of participant were not included since these variable only include married and employed participant respectively. Other variables like Sex, educational status, drug use, during the previous 12 months, residence were additionally considered

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using a p<0.20 screening threshold and clinical importance. Each candidate variable was added separately to the initial model, and models were compared using Akaike's information criterion (AIC). The model with drug use variable had the lowest AIC and was selected as the preferred base model. Sex was subsequently retained in the final model based on its epidemiological importance from previous literature. Participant occupation and perceived spouse support were excluded from the primary model because they were applicable only to employed and married participants, respectively. Adjusted odds ratios (AORs) with 95% confidence intervals (CIs) were reported. Multicollinearity, discrimination, and calibration were assessed using GVIF, the area under the receiver operating characteristic curve (AUC), and the Hosmer–Lemeshow goodness-of-fit test, respectively. Firth's penalized logistic regression was performed as a sensitivity analysis to

Ethical consideration

Ethical approval was obtained from the Nepal Health Research Council (Ref. 12/2019) and a official institutional approval from PAHS was also obtained before the study. During before interview, an informed written consent was obtained from each study participant. All the participants written informed consent was obtained, and participation was voluntary with the right to withdraw at any time without consequence

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Descriptive Summary

Table 1: Demographic Characteristics of Participants

Characteristic	**N = 256**[1]
Sex	
Male	115 (45%)
Female	141 (55%)
Age of participant	40 (34, 49)
Age Group	
< 49	191 (75%)
49 ≥	65 (25%)
Residence	
Rural	40 (16%)
Urban	216 (84%)
Ethnicity	
Dalit	53 (21%)
Non-Dalit	203 (79%)
Religion	
Hindu	197 (77%)
Buddhist	39 (15%)
Muslim	1 (0.4%)
Christian	19 (7.4%)
Marital Status	
Married	165 (64%)
Unmarried	29 (11%)
Divorced/separated/widowed	62 (24%)
Family Type	
Joint	99 (39%)
Nuclear	157 (61%)
Educational Status	
Formal Education	179 (70%)
Illiterate/Informal	77 (30%)

[1]n (%); Median (Q1, Q3)

Table 1 shows among the 278 participants recruited, 22 were non-respondents, resulting in a non-response rate of approximately 8%. Among the 256 respondents who included in the study, 55.0% (n = 141) were female. The median age of the participants was 40 years (Q1 = 34, Q3 = 49). More than three-quarters of the participants 191 (75%) were aged below 49 years. The majority of the respondents 84.4% (n = 216) resided in urban areas since ART centre was in city. More than three-quarters of the participants 79.3% (n = 203) belonged to non-Dalit caste groups, whereas 20.7% (n = 53) were Dalit. The majority of the participants were Hindu 77.0% (n = 197). In marital status, nearly two-thirds of the participants 64.5% (n = 165) were married followed by 24.3% *Divorced/separated/widowed* status. More than three-fifths of the participants 61.3% (n = 157) lived in nuclear families. Regarding educational status, approximately one-third of the participants 30.1% (n = 77) were illiterate or had only informal education.

Table 2: Economic Characteristics of Participants

Characteristic	**N = 256**[1]
Employment in 12 months	
Unemployed	75 (29%)
Employed	181 (71%)
Income (NRP)	15000 (10000, 20000)
Monthly Income Categories (NRP)	
≤ 10,000	139 (54%)
>10,000	117 (46%)
Occupation (n = 181)	
Business	52 (29%)
Agriculture	40 (22%)
Labor	46 (25%)
Service	17 (9.4%)
Driving	11 (6.1%)
Other	15 (8.3%)
Family Occupation	
Business	55 (21%)
Agriculture	52 (20%)

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Characteristic	N = 256[1]
Labor	49 (19%)
Civil Service	21 (8.2%)
Foreign employment	30 (12%)
Pension	16 (6.3%)
Driving	9 (3.5%)
Other	24 (9.4%)
Family Monthly Income (NRP)	
10,000 below	24 (9.4%)
10,000 to 50,000	216 (84%)
>50,000	16 (6.3%)

[1]n (%); Median (Q1, Q3), Nepali Rupees Paisa (NRP)

Table 2 shows among total participant, 71% (n= 181) of participant were employed, the median (Q2) of participant monthly income was 15,000 Nepali Rupees Paisa (NRP), Q1 was 10000 NRP and Q3 was 20,000 NRP. Approximately 54% of participants had a monthly income below or equal to Q1 cut-off (NRP 10,000). Among employed participant, majority had occupation of Business 29% (n= 52) followed by Labor 25% (n= 46). Similarly, the most common family occupation was Business 21% (n=55) followed by Agriculture 20% (n=50). Regarding monthly family income, the majority of participant 84%, (*n* = 216) reported an income between NRP 10,000 and 50,000 and followed by 9.4% (*n* = 24) with a below NRP 10,000, while 6.3% (*n* = 16) reported a income of more than NRP 50,000.

Table 3: Socio-behavioural character of Participants

Characteristic	N = 256[1]
Family Support	
Yes	28 (11%)
No	228 (89%)
Spouse Support (n=165)	
Yes	6 (3.6%)
No	159 (96%)
Involvement in family events	
Yes	18 (7%)
No	238 (93%)
Perceived Discrimination	
Yes	57 (22%)
No	199 (78%)
Current Tobacco Consumption	
No tobacco consumption	167 (65%)
Cigarette Smoking	37 (15%)
Smokeless tobacco consumption	45 (18%)
Both	7 (2.7%)
Current Alcohol Use	
Yes	57 (22%)
No	199 (78%)
Drugs in past 12 months	
Yes	10 (3.9%)
No	246 (96%)
Medication of Depression	
Yes	14 (5.5%)
No	242 (95%)

[1]n (%); Median (Q1, Q3),Nepali Rupees Paisa (NRP), K (Thousand)

Table 3 shows only 11% of participants (*n* = 28) reported perceiving family support, whereas the majority (89%, *n* = 228) reported not perceiving family support. Similarly, among married participant (n=165), only 3.6% (n=6) reported perceived support from spouse whereas most 96% (n=159) participant reported no perceived support from spouse. Only 7% (n=18) participant reported involvement in family events whereas 93% (n=238) not reported involvement in family events. On the other hand, 22% (n =57) participant reported perceived discrimination in society while the remaining 78% (n=199) did not. Majority of Participant 65% (n= 166) report not consuming current tobacco, whereas 18% (n= 45) were smokers and 15% (n= 37) used smokeless tobacco and 2.7% (n= 7) consumed both smoked and smokeless tobacco products. Most participants 78% (*n* = 199) were not current alcohol consumers. Similarly, the vast majority of participants 96% (*n* = 246) reported not using any addictive drugs during the past 12 months, while only 3.9% (*n* = 10) reported drug use during 12 months. And only 5.5% of participants (*n* = 14) reported currently taking medication for depression.

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Table 4: Depressive Symptoms and categories in study participants

Characteristic	N = 256[1]
Depressive Symptoms (Yes/No)	
Yes	49 (19.1%)
No	207 (80.9%)
Median-BDI Score	8 (4, 14)

[1]n (%), median (Q1, Q3)

Table 4 presents the prevalence of depressive symptoms among the study participants. Of the total 256 participants, 19.1% (95% CI: 14.8–24.4%) (n = 49) had depressive symptoms, while 80.9% (n = 207) did not have depressive symptoms. The median BDI score of total participant was 8 with Q1 was 4 and Q3 was 14.

Table 5:Test of Association using Pearson's Chi-squared test; Fisher's exact test between different variable and depressive symptoms

Characteristic	No N = 207[1]	Yes N = 49[1]	p-value[2]
Sex			0.056
Male	87 (42%)	28 (57%)	
Female	120 (58%)	21 (43%)	
Age group			0.9
< 49	154 (74%)	37 (76%)	
≥ 49	53 (26%)	12 (24%)	
Residence			0.14
Rural	29 (14%)	11 (22%)	
Urban	178 (86%)	38 (78%)	
Ethnicity			0.002*
Dalit	35 (17%)	18 (37%)	
Non-Dalit	172 (83%)	31 (63%)	
Marital Status			0.001*
Married	144 (70%)	21 (43%)	
Unmarried	22 (11%)	7 (14%)	
Divorced/separated/widowed	41 (20%)	21 (43%)	
Educational Status			0.14
Formal Education	149 (72%)	30 (61%)	
Illiterate/Informal	58 (28%)	19 (39%)	
Employment status			0.4
Un-Employed	58 (28%)	17 (35%)	
Employed	149 (72%)	32 (65%)	
Monthly Income (NRP)			0.041*
>10,000	101 (49%)	16 (33%)	
≤ 10,000	106 (51%)	33 (67%)	
Occupation			0.002*
Non-Labor	118 (79%)	17 (53%)	
Labor	31 (21%)	15 (47%)	
Family Income (NRP)			0.20
10K below	16 (7.7%)	8 (16%)	
10K to 50K	177 (86%)	39 (80%)	
More than 50K	14 (6.8%)	2 (4.1%)	
Family Occupation			0.023*
Non-Labor	173 (84%)	34 (69%)	
Labor	34 (16%)	15 (31%)	
Current Alcohol Use			>0.9
Yes	46 (22%)	11 (22%)	
No	161 (78%)	38 (78%)	
Drug Use in 12 months			0.10
Yes	6 (2.9%)	4 (8.2%)	
No	201 (97%)	45 (92%)	
Tobacco Consumption			<0.001*
Yes	62 (30%)	28 (57%)	
No	145 (70%)	21 (43%)	
Family Support			0.7
Yes	22 (11%)	6 (12%)	

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Characteristic	No N = 207[1]	Yes N = 49[1]	p-value[2]
No	185 (89%)	43 (88%)	
Spouse Support			0.028*
Yes	3 (2.1%)	3 (14%)	
No	141 (98%)	18 (86%)	
Involvement Family Events			0.4
Yes	13 (6.3%)	5 (10%)	
No	194 (94%)	44 (90%)	
Discrimination in Society			<0.001*
Yes	35 (17%)	22 (45%)	
No	172 (83%)	27 (55%)	

[1]n (%)
[2]Pearson's Chi-squared test; Fisher's exact test, **P value <0.05*

Table 5 showed test of association by chi-square test and fisher's exact test, we found that variable Ethnicity (p = 0.002), Marital status (p = 0.001), Monthly Income (p = 0.041), Occupation of participant (p = 0.002),Family Occupation (p = 0.023), Current Tobacco Consumption (p = 0.001), Perceived Spouse Support (p=0.028) and Perceived Discrimination in society (p = 0.001). However other factor like Sex (p = 0.056), Age group (p = 0.9), Educational Status (p = 0.14), Residence (p = 0.14), Employment Status (p = 0.4), Family income (p = 0.20), Current use (p = 0.9), Drug Use in 12 months (p = 0.10), Family Support (p = 0.7) and Involvement in Family events (p = 0.4) were not statistically significant from Pearson's Chi-squared test or Fisher's exact test

Table 6: Multivariable logistic regression

	Crude OR (95% CI)			Adjusted OR (95% CI)		
Variable	**OR**	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**
Sex						
Female	—	—		—	—	
Male	1.84	0.98, 3.48	0.058	1.87	0.76, 4.73	0.20
Ethnicity						
Non-Dalit	—	—		—	—	
Dalit	2.85	1.42, 5.64	0.003	2.00	0.89, 4.42	0.089
Marital Status						
Married	—	—		—	—	
Unmarried	2.18	0.78, 5.56	0.11	2.54	0.77, 7.86	0.11
Divorced/Separated/Widowed	3.51	1.75, 7.10	<0.001	3.05	1.29, 7.35	0.012*
Monthly Income						
>10000	—	—		—	—	
≤ 10,000	1.97	1.03, 3.87	0.044	2.89	1.24, 7.12	0.017*
Family Occupation						
Non-Labor	—	—		—	—	
Labor	2.24	1.08, 4.52	0.026	1.85	0.78, 4.32	0.20
Perceived Discrimination Society						
No	—	—		—	—	
Yes	4.00	2.04, 7.85	<0.001	2.49	1.14, 5.40	0.021*
Drug use in 12 months						
No	—	—		—	—	
Yes	2.98	0.74, 10.9	0.10	3.65	0.75, 16.5	0.094
Current Tobacco Consumption						
No	—	—		—	—	
Yes	3.12	1.65, 5.97	<0.001	4.84	2.04, 12.1	<0.001*

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Table 6 showed multivariable logistic regression model, Divorced/separated/widowed participants had 3.05 times significantly higher odds of the depressive symptoms than married participants (AOR = 3.05, 95% CI: 1.29-7.35, p = 0.012), while the odds of outcomes were 2.54 times higher for Un-married participant (AOR = 2.54, 95% CI: 0.77-7.86, p = 0.11) than married however was not statistically significant. For monthly income, ≤10,000 NRP had 2.89 fold higher odds of depressive symptoms than those with an income >10,000 NRP (AOR: 2.89, 95% CI: 1.24−7.12, p = 0.017) of outcomes than those who had monthly income above 10,000. Participants who had perceived discrimination had two and half times (AOR: 2.49, 95%

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CI:1.14-5.40, p =0.021) higher odds of depressive symptoms than those did not reported perceiving discrimination. Among Tobacco consumption, participant who had consumed current tobacco product were 4.84 times (AOR: 4.84, 95% CI: 2.04-12.1, p =0.001) had higher odds of depressive symptoms than those did not reported consuming tobacco product currently.

Male participant had 1.87 times higher the odds of depressive symptoms than female (AOR:1.87, 95% CI: 0.76-4.73, p = 0.20).Participant from Dalit ethnicity had 2 times higher odds depressive symptoms than Non-Dalit (AOR 2.0, 95%,CI: 0.89-4.42, p = 0.089). Participant belonging from family with labor occupation had 1.85 times higher odds of depressive symptoms than non-labor family occupation (AOR: 1.85, 95% CI : 0.78-4.32, p = 0.20). Participant reporting drug use with in 12 months had 3.65 times higher odds of depressive symptoms than who did not (AOR: 3.65, 95% CI: 0.75, 16.5, p =0.094).However these odd ratio were not statistically significant after adjustment.

The final multivariable logistic regression model demonstrated good discrimination (AUC = 0.814, 95% CI 0.749–0.879) and adequate calibration (Hosmer–Lemeshow $\chi^2 = 8.28$, df = 8, p = 0.4069). And there is no evidence of multicollinearity since GVIF-adjusted values below 1.28. Sensitivity analysis using Firth's penalized logistic regression yielded broadly similar estimates to the conventional logistic regression model. Divorced/separated/widowed marital status (OR 2.85, 95% CI 1.24–6.69), income *≤ 10,000* NRP (OR 2.68, 95% CI 1.18–6.38), tobacco consumption (OR 4.39, 95% CI 1.91–10.61), and perceived discrimination (OR 2.38, 95% CI 1.12–5.03) remained statistically associated with depression. Drug use remained positively associated but was imprecisely estimated (OR 3.50, 95% CI 0.79–14.59; p=0.096). Overall, the direction and magnitude of associations were consistent with the primary logistic regression analysis.

Discussion

Out of 278 study sample 22 did not responded resulting in total 256 participant. Most of participant were under 49 age and and the distribution by sex was relatively equal. The prevalence of depressive symptoms in PLWH in Pokhara valley was 19% , this result is very low than result of global meta-analysis pooled prevalence of depressive symptoms in PLWH around the world 31% (95% CI 28% to 34%), including highest prevalence 40% from South-East Asia, another pooled prevalence 39.1% (95% CI: 30.4 to 48.5) in India, however both pooled estimates had very high heterogeneity .

Our study results were similar to prevalence 25.5 % – 29.3 % by reported by 2 studies done in Kathmandu Valley in 2010 (6,10). However, another study in Kathmandu, Lalitpur and Kavre district had very high prevalence 40% (16). Another study in Lumbini province reported the prevalence of depressive symptoms in PLWH as 26.8% (10), however lower cut-off ≥13 was used reducing the specificity of result (15). The prevalence of depressive symptoms in general population is found very low around 6.2%, suggesting PLWH are higher risk compared to general (10). The Nepal's National Mental Health Survey (2019/20) also reported a 1% prevalence of major depressive at national level and 0.6 % in Gandaki Province although, this difference should be interpreted cautiously due to differences in study populations and assessment methods, as the national survey used a diagnostic interview while our study used a screening tool (3).

In general in Nepali population, married population has lower odds of mental health disorder including depressive symptoms than higher odds at divorced/widowed or separated person (18,19). Similar pattern was also seen in PLWH in our study, where Divorced/separated/widowed participants had 3.05 times higher odds of the depressive symptoms than married participants (AOR = 3.05, 95% CI: 1.29-7.35, $p = 0.012$). However, this variable was not significant in other studies (6,10,13).

Income is a common socio-economic determinant of depressive symptoms, which was also reported in a study conducted in Lumbini Province, where participants in the lowest income percentile had higher odds of depressive symptoms compared with those in the highest income percentile (13). Similarly, participants with a monthly income ≤ 10,000 NRP (the first -quartile of cut-off) had nearly three times higher odds of depressive symptoms than those earning >10,000 NRP (AOR: 2.89, 95% CI: 1.24–7.12, p = 0.017). This finding is also consistent with meta-analysis conducted in Ethiopia (20).

HIV-related stigma and discrimination remain prevalent across Nepalese society and may contribute to low self-esteem, depressive symptoms, and social isolation (23). Our study 22.2 % (n=57) of participant reported perceived discrimination in society, those who perceived had around 2.5 times higher odds of depressive symptoms than those who did not (AOR: 2.49,

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95% CI:1.14-5.40, p =0.021). This results was similar in South African study where perceived discrimination and internal stigma were associated with occurrence of depressive symptoms (21).

Our study showed that participant who were current tobacco product consumer had 4.84 times (AOR: 4.84, 95% CI: 2.04-12.1, p =0.001) had higher odds of depressive symptoms than those who did not. Similar findings was reporting significant association between tobacco use (including smoking or smokeless tobacco) and depressive symptoms among PLWH were evident in Kathmandu Valley and South Africa (13,22).

Dalit ethnicity ranks least at bottom of hierarchal caste system in Nepal (23). It was evident that they have socioeconomic disadvantage with increased risk of mental health problems (19), however ethnicity variable was not statistically associated with depressive symptoms in our study (AOR 2.0, 95%,CI: 0.89-4.42, p = 0.089) and this finding was consistent in previous study (AOR:1.72, 95% CI: 0.42-7.16) (10). Although previous study reported female sex was associated with depressive symptoms than male (24), however our study (AOR:1.87, 95% CI: 0.76-4.73, p = 0.20) and other studies could not find statistically associated in depressive symptoms and Sex in PLWH (12, 25). Similarly, a previous study from Lumbini province Nepal found that unemployed participant higher odds (3.28, AOR 1.63–6.60) than self-employed (13). A Kathmandu study reported participant who had used drug within past 6 month had significant higher odds of depressive symptoms (6). However in our study these variables including participant employment, occupation and drug use in past 12 month were not statistical significant.

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Strengths and Limitations

The major limitation of this study is clinical data were not collected considering difficulties of ethical aspect of ART centre and patient data privacy. Participants were recruited from a single ART centre using institutional based consecutive sampling. Although all eligible individuals attending the centre during the study period were approached, people who did not attend the ART centre during this period were not represented. Therefore, the findings may not be generalizable to all PLWH in Kaski District, particularly those who do not take regular service from the ART. The cross-sectional study design limits to explain the causal inference. Major of this study strength is the study used valid version of BDI tool for depression assessment.

Conclusion

This study concludes that 19.1% of the PLWH in ART centre had depressive symptoms using BDI tool and Marital Status, Monthly income, Perceived discrimination and Tobacco consumption were associated with depressive symptoms in PLWH. The study highlights burden of depressive symptoms in PLWH and we recommend policy maker and planner to intergrate mental health services into routine ART services. As this was a single-centre cross-sectional study with a limited sample, findings should be interpreted cautiously and may not be generalisable to all PLWH in Nepal. Multicentre longitudinal studies are needed to further investigate the burden and determinants of depressive symptoms.

Patient and Public Involvement:

There was no Patient and Public involvement

Funding:

There was no source of fund for this study.

Data Statement:

All data, code and study materials are available at author github repo https://github.com/ragathefreeman/Depression-among-PLHV-in-Nepal

Conflict of Interest:

The authors declare that they have no conflict of interest.

Author contributions

Khanal R: Conceptual Framework, Ethical Approval, Tool development, Protocol writing, Data collection and analysis, Manuscript Writing, Administration and management, Data sharing.

K.C A: Conceptual Framework, Tool development, Protocol writing, Data collection and analysis, Review and supervision.

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Supplement A: STORB Reporting Guideline for observational studies(17).

	Item Description	Location (or reason for not reporting)
Title and abstract		
1a. Indicate the study's design	Indicate the study's design with a commonly used term in the title or the abstract.	2
1b. Abstract	Provide in the abstract an informative and balanced summary of what was done and what was found.	2

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Introduction		
2. Background / rationale	Explain the scientific background and rationale for the investigation being reported.	3
3. Objectives	State specific objectives, including any prespecified hypotheses.	3
Methods		
4. Study design	Present key elements of study design early in the paper.	4
5. Setting	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection.	4
6a. Eligibility criteria	**Cohort study:** Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up. **Case-control study:** Give the eligibility criteria, and the sources and methods of case ascertainment and control selection. Give the rationale for the choice of cases and controls. **Cross-sectional study:** Give the eligibility criteria, and the sources and methods of selection of participants.	4
6b. Matching criteria	**Cohort study:** For matched studies, give matching criteria and number of exposed and unexposed. **Case-control study:** For matched studies, give matching criteria and the number of controls per case.	
7. Variables	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable.	5
8. Data sources / measurement	For each variable of interest give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group.	5
9. Bias	Describe any efforts to address potential sources of bias.	5
10. Study size	Explain how the study size was arrived at.	4
11. Quantitative variables	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen, and why.	5
12a. Statistical methods	Describe all statistical methods, including those used to control for confounding.	5
12b. Statistical methods – subgroups and interactions	Describe any methods used to examine subgroups and interactions.	5
12c. Statistical methods – missing data	Explain how missing data were addressed.	
12di. Statistical methods – loss to follow-up	**Cohort study:** If applicable, describe how loss to follow-up was addressed.	
12dii. Statistical methods – matching cases and controls	**Case-control study:** If applicable, explain how matching of cases and controls was addressed.	

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12diii. Statistical methods – sampling strategy	**Cross-sectional study:** If applicable, describe analytical methods taking account of sampling strategy.	5
12e. Statistical methods – sensitivity analyses	Describe any sensitivity analyses.	5,6
Results		
13a. Participant numbers	Report the numbers of individuals at each stage of the study—e.g., numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed; Consider use of a flow diagram.	7
13b. Participants – non-participation	Give reasons for non-participation at each stage.	7
13c. Participants – flow diagram	Consider use of a flow diagram.	
14a. Descriptive data – participant characteristics	Give characteristics of study participants (e.g., demographic, clinical, social) and information on exposures and potential confounders. Present the information in a table.	7, 8
14b. Descriptive data – missing data	Indicate the number of participants with missing data for each variable of interest.	
14c. Descriptive data – follow-up time	**Cohort study:** Summarise follow-up time—e.g., average and total amount.	
15. Outcome data	**Cohort study:** Report numbers of outcome events or summary measures over time. **Case-control study:** Report numbers in each exposure category, or summary measures of exposure. **Cross-sectional study:** Report numbers of outcome events or summary measures.	8,9
16a. Main results	Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (e.g., 95% confidence intervals). Make clear which confounders were adjusted for and why they were included.	8,9,10
16b. Main results – category boundaries	Report category boundaries when continuous variables were categorised.	8,9,10
16c. Main results – risk	If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period.	
17. Other analyses	Report other analyses done—e.g., analyses of subgroups and interactions, and sensitivity analyses.	
Discussion		
18. Key results	Summarise key results with reference to study objectives.	11
19. Limitations	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias.	11
20. Interpretation	Give a cautious overall interpretation considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence.	11

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21. Generalisability	Discuss the generalisability (external validity) of the study results.	11,12
Other information		
22. Funding	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based.	12

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